Comparative snRNA-seq analysis of control and multiple sclerosis brain tissue

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Multiple sclerosis: a chronic inflammatory CNS disease

- Progressive neuroinflammatory autoimmune disease
- Characterized by:
  - Demyelination
  - Neuroinflammation
  - Neurodegeneration
- Leading cause of disability in young adults
- Focal inflammatory lesions

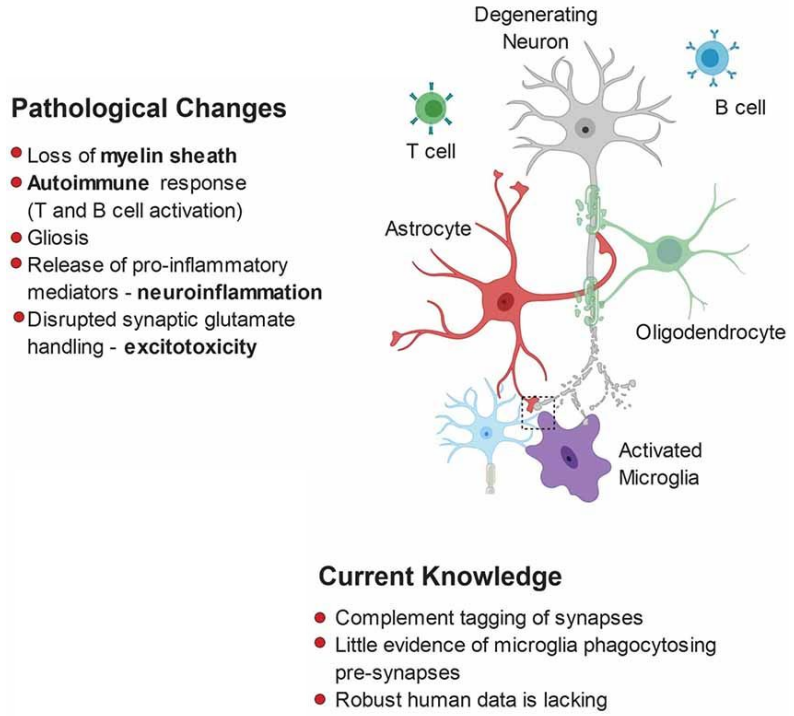
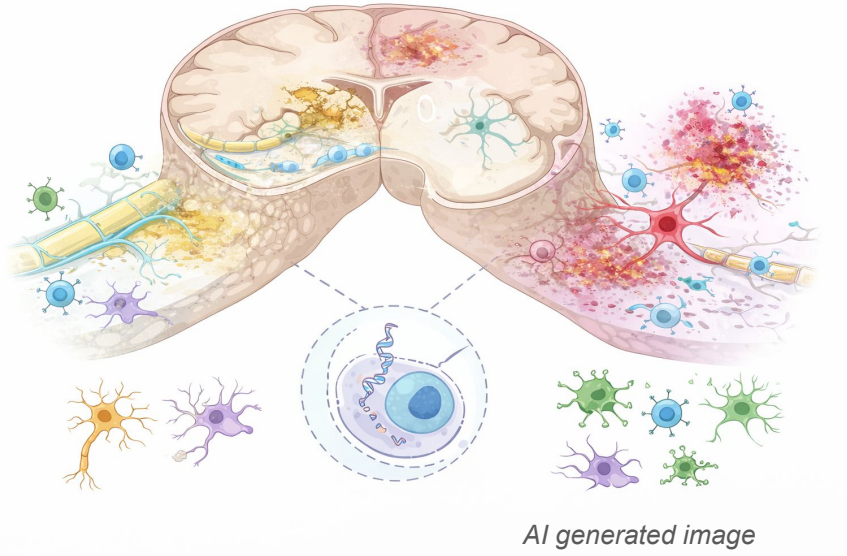


Figure from: Henstridge et al., *Front. Cell. Neurosci.* (2019)

Why a single-nucleus approach in MS?

- MS lesions show strong **regional heterogeneity**
- The brain consists of **highly diverse cell populations**, each with distinct functions
- Cells can exist in **multiple activation states**, which may reflect **disease progression**
- Single-nucleus RNA-seq allows:
  - Cell-type–specific resolution
  - Detection of disease-associated transcriptional states



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Key Open Questions

- ***Does MS affect all cortical neurons equally?***
- ***Are there any cell-type specific disease progression mechanisms?***

The dataset:

HUMAN CELL ATLAS
DATA PORTAL

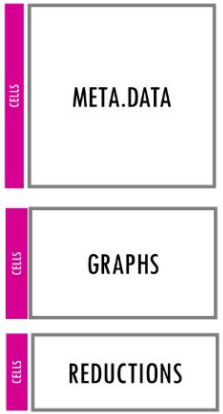
10x v2 single-nucleus RNA-seq from human brain cells

Subset (12 samples):

- **MS:** 3 male + 3 female
- **Control:** 3 male + 3 female

QC and Filtering

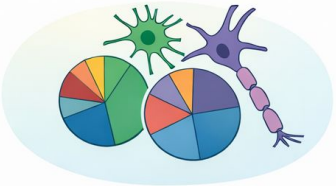




Cell Type Assignment

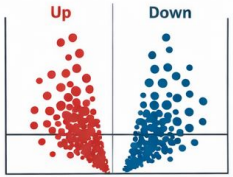
Cell-Type Composition Analysis (DCATS)

Cell type differences in MS vs Control?



Differential Gene Expression (DESeq2)

Cell-specific gene changes?



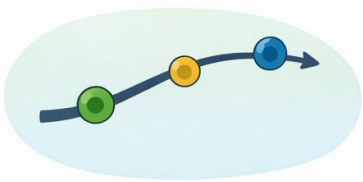
Pathway-Level Analysis (*fgsea*)

Affected biological pathways?



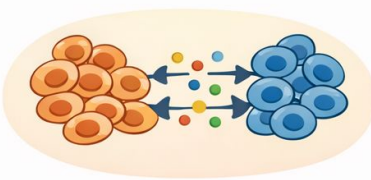
Trajectory Analysis (*Slingshot*)

Cellular state transitions?



Cell-Cell Communication (CCC)

Altered cell signaling?



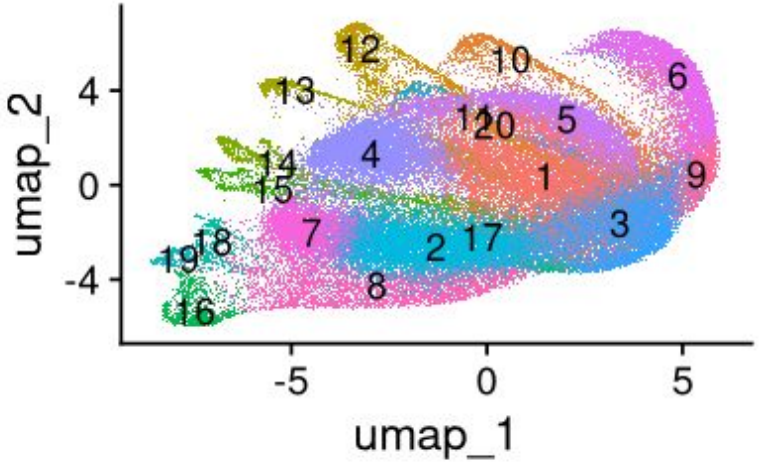
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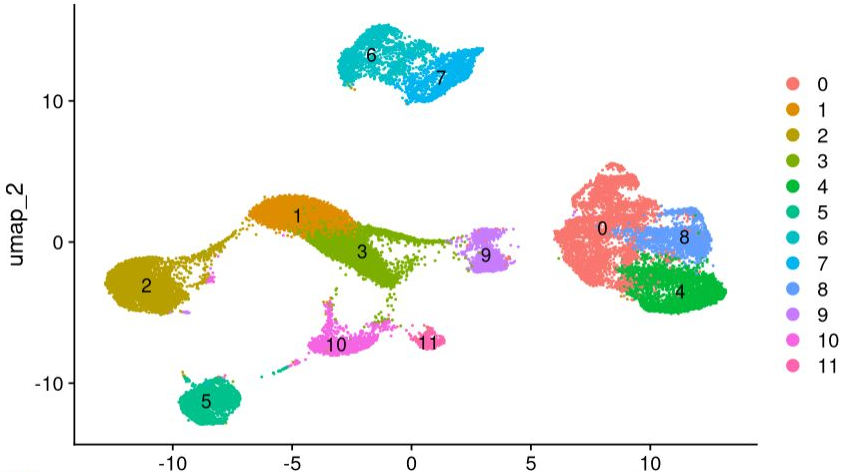
QC and Filtering

- Computed QC metrics (%mito, %ribo) and visualized the distribution using violin plots.
- Established thresholds using plots and percentiles.
- Filtering goal:
  - Remove empty droplets
  - Remove doublets
  - Remove damaged nuclei (high %mito)
- Applied looser thresholds due to the sample type (higher RNA degradation, higher ambient contamination, tissue processing is harder)
- Removed around 32% of nuclei
- Pre-processing: log-normalization + scaling (regression %mito)

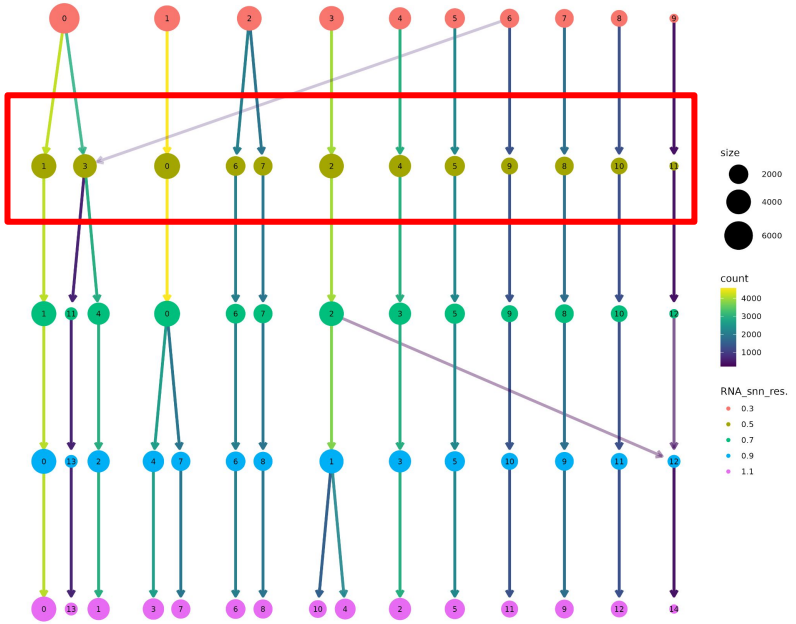
QC and Filtering

- Due to the poor separation of clusters and the impossibility of cell annotation, we had to apply stricter filtering as they do in the paper.



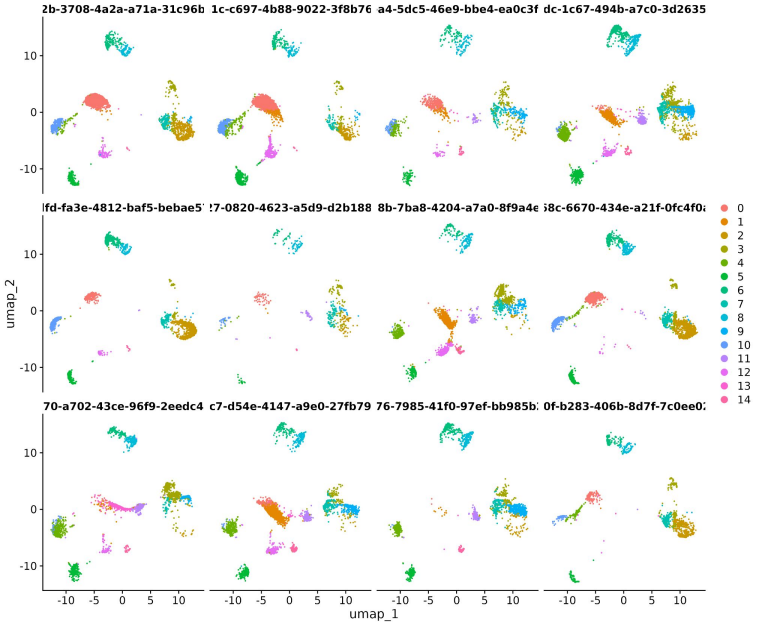


Clustering Resolution Selection & Integration Assessment



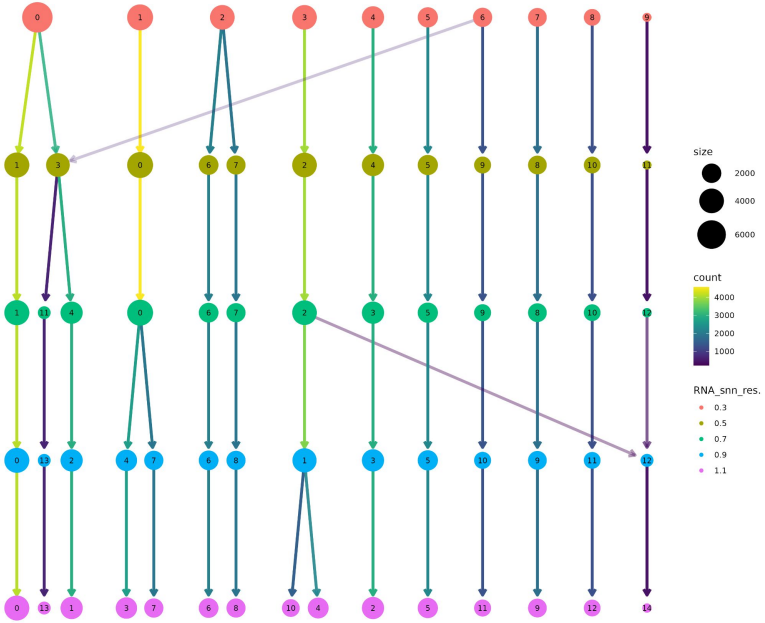
- **"*Louvain*" Clustering**
- **Optimal Resolution:** 0.5

Sample:



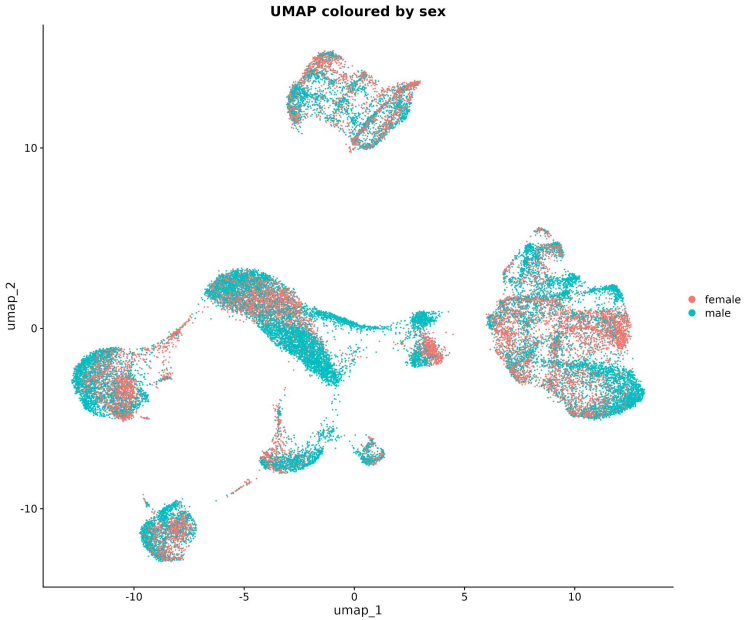
- Clustering is driven by **biological cell states**, not technical variation

Clustering Resolution Selection & Integration Assessment



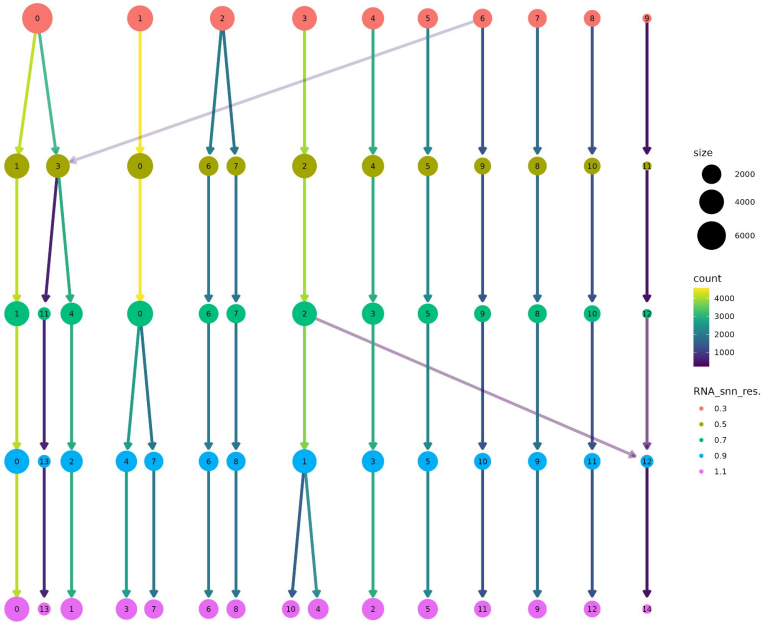
- **"*Louvain*" Clustering**
- **Optimal Resolution:** 0.5

Sex:



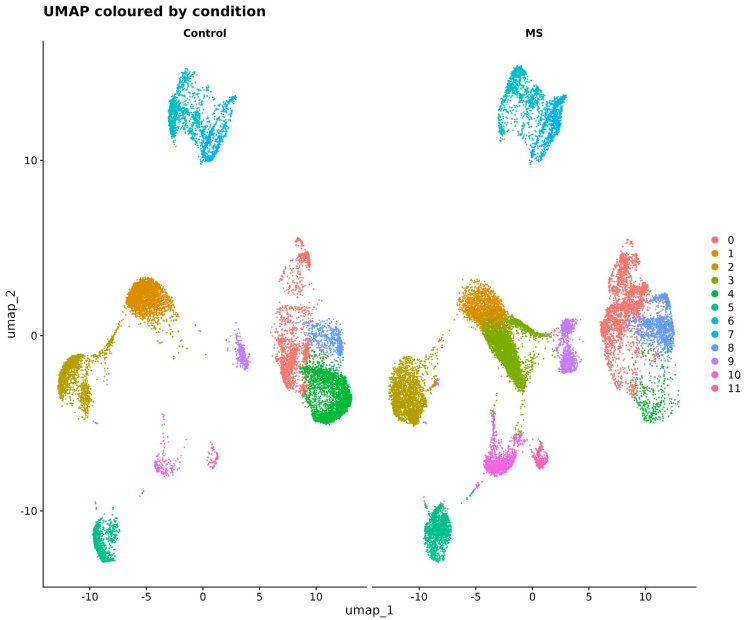
- Clustering is driven by **biological cell states**, not technical variation

Clustering Resolution Selection & Integration Assessment



- **"*Louvain*" Clustering**
- **Optimal Resolution:** 0.5

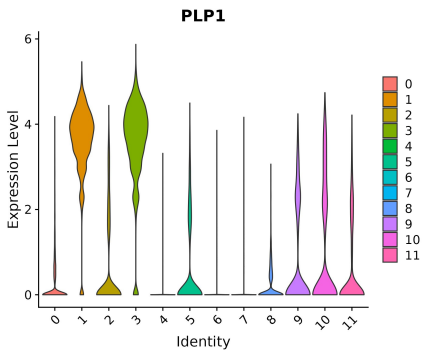
Condition:

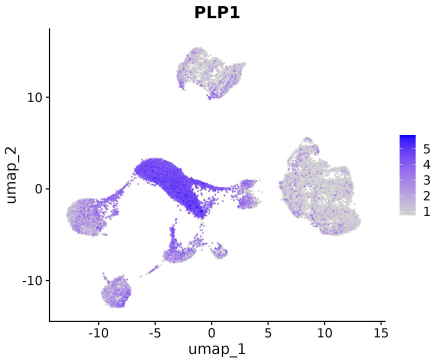


- Clustering is driven by **biological cell states**, not technical variation

Cell Type Assignment

List of gene markers



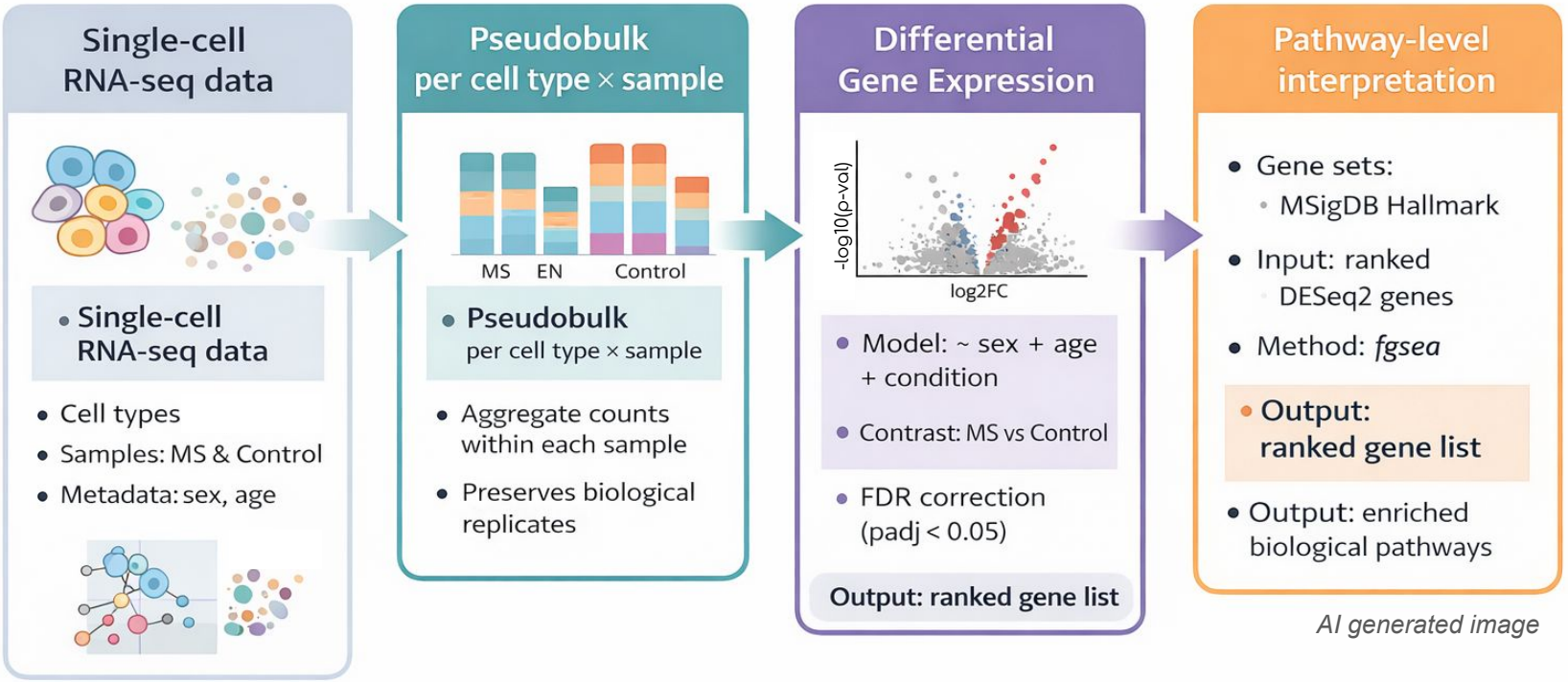


`FindConservedMarkers()`

SLC1A3	DOCK8
ADGRV1	FYB1
RNF219-AS1	APBB1IP
SLC1A2	TBXAS1
RYR3	PTPRC
NEAT1	C3

- Selection of **established cell-type markers** based on literature and the reference paper.
- **Marker expression assessment across clusters** using violin plots.
- **UMAP-based visualization** of marker distribution to support cluster identification.
- Identification of ambiguous clusters showing **low or non-specific marker expression.**
- Assignment of cell types based on **cluster-specific conserved gene signatures.**

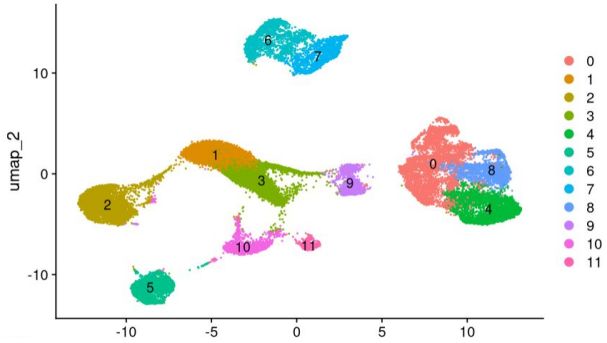
Differential Gene Expression and Gene-Set Enrichment Analysis

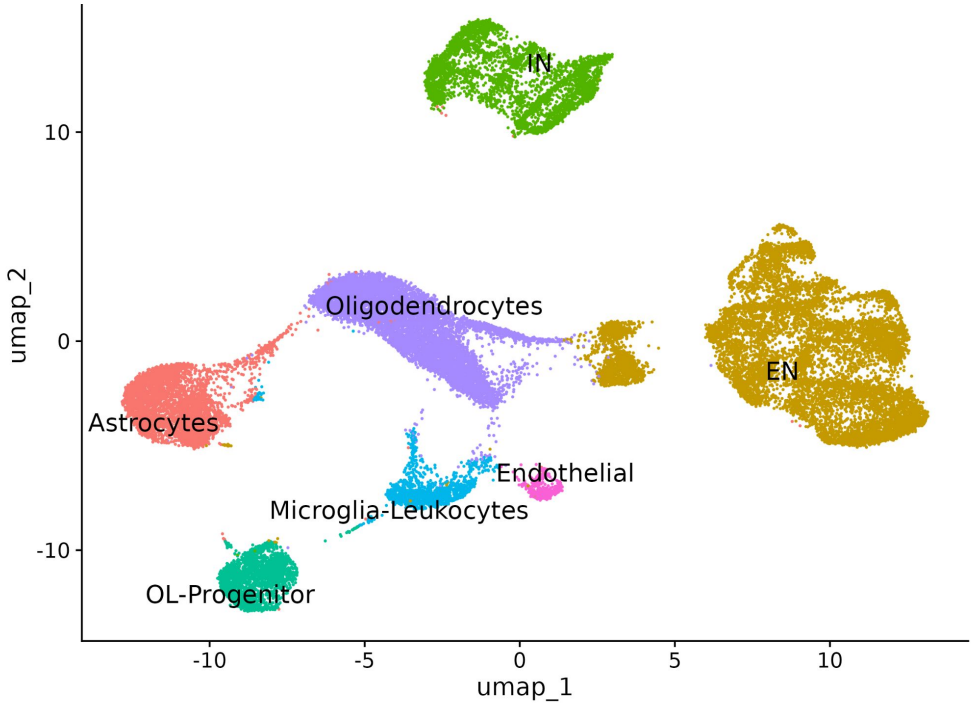


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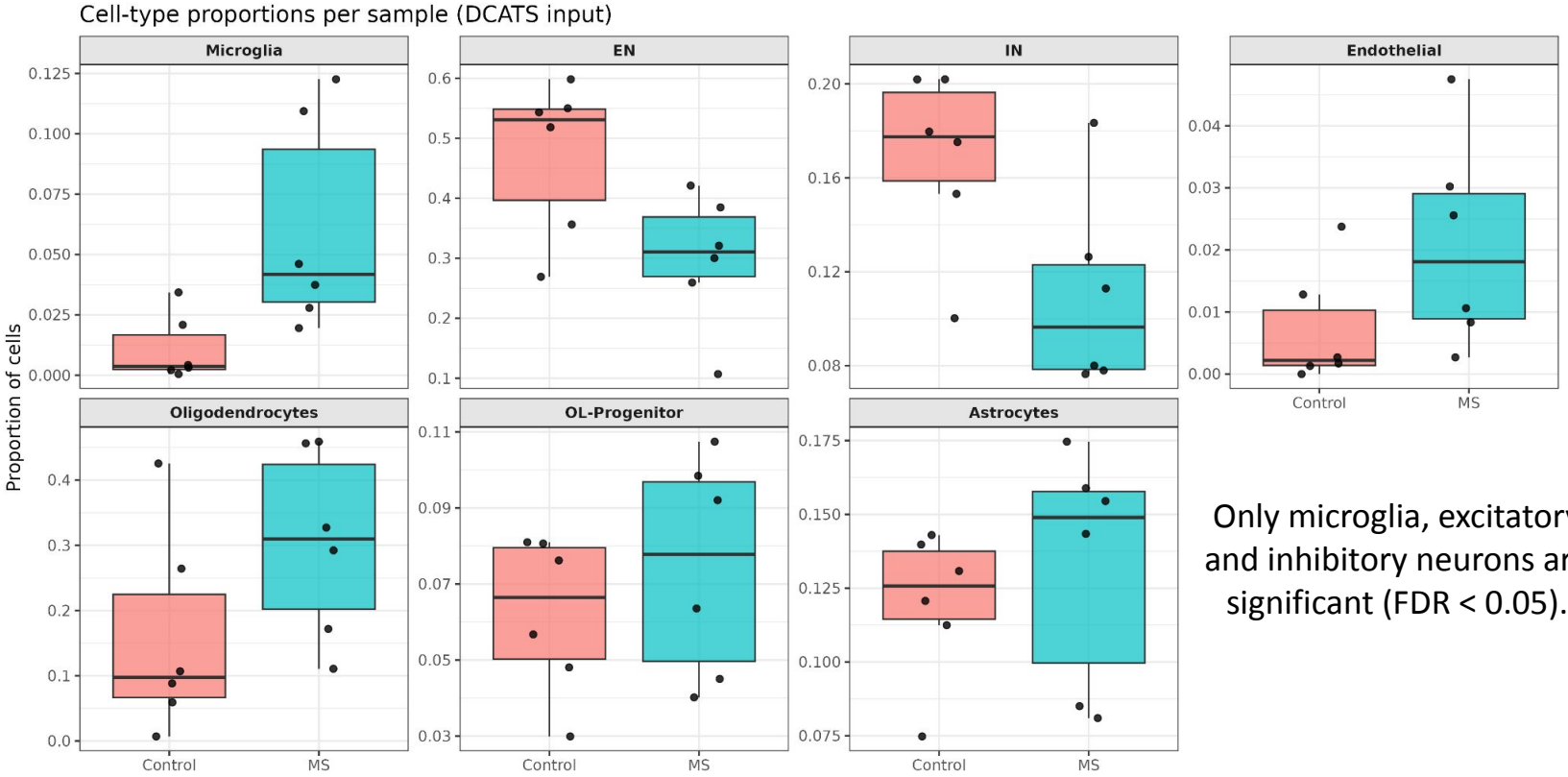
Cell Type Assignment

Assign a cell type to each cluster.





Differential Composition Analysis

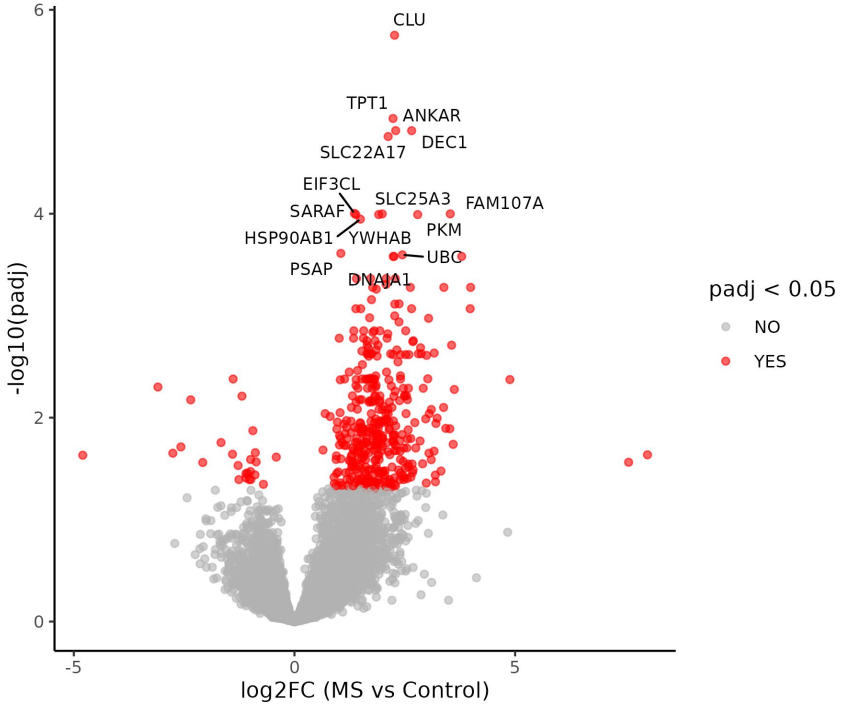


Only microglia, excitatory and inhibitory neurons are significant (FDR < 0.05).

Differential Gene Expression Analysis

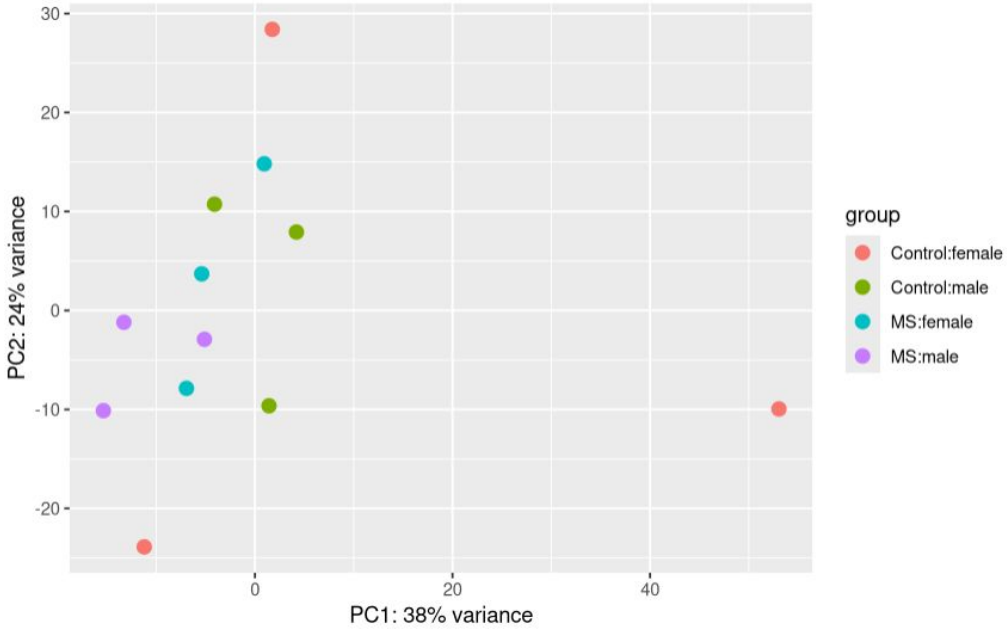
- **Excitatory neurons:** Drastic differential expression with several hundred genes significantly altered between MS and control samples.
  - Stress response, protein homeostasis, and immune-related genes
- **Oligodendrocytes:** Strong differential expression signal, comparable in magnitude to EN.
  - Metabolic, mitochondrial, and stress-response genes
- **Inhibitory neurons:** Detectable differential expression
  - Heat-shock response genes and long-noncoding RNAs
- **OL progenitors:** Weak but detectable differential expression
  - Heat-shock response genes
- **Astrocytes:** Weak differential expression
  - Long-noncoding (lnc) RNAs found, regulated in response to inflammation and stress
- **Microglia, endothelial cells:** No significant differential expression after multiple testing correction.

Volcano Plot for Oligodendrocytes:



Interpreting weak differential expression signals

- Limited condition separation in *Pseudobulk* PCA
- *Pseudobulk* averages heterogeneous cell states
- Subsetting reduces statistical power
- MS effects may be compositional, not transcriptional

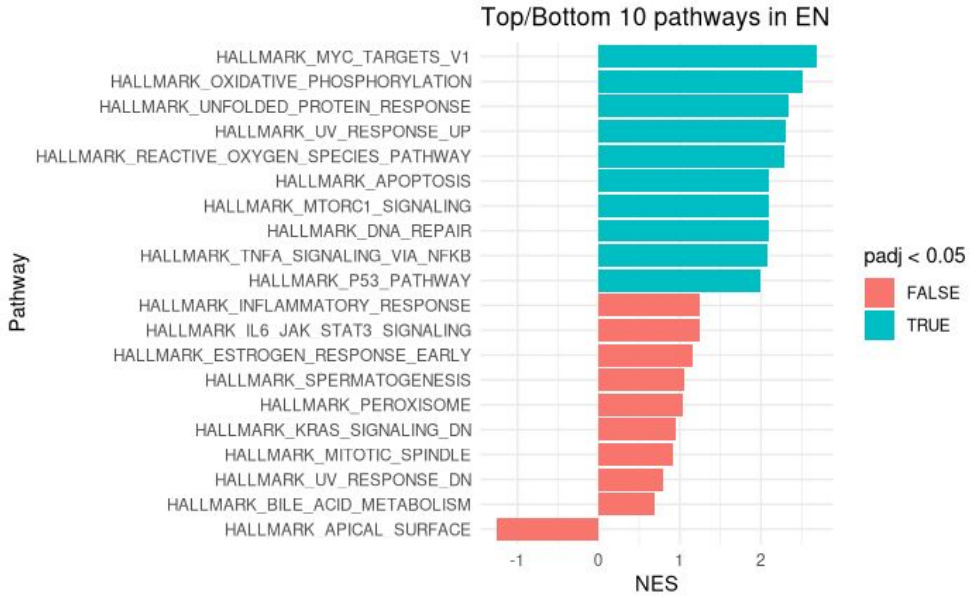


Gene-set enrichment analysis

Excitatory Neurons (EN)

Enriched pathways in MS:

- Oxidative phosphorylation
- MYC targets
- Unfolded protein response
- Reactive oxygen species pathway
- mTORC1 signaling
- Apoptosis / DNA repair

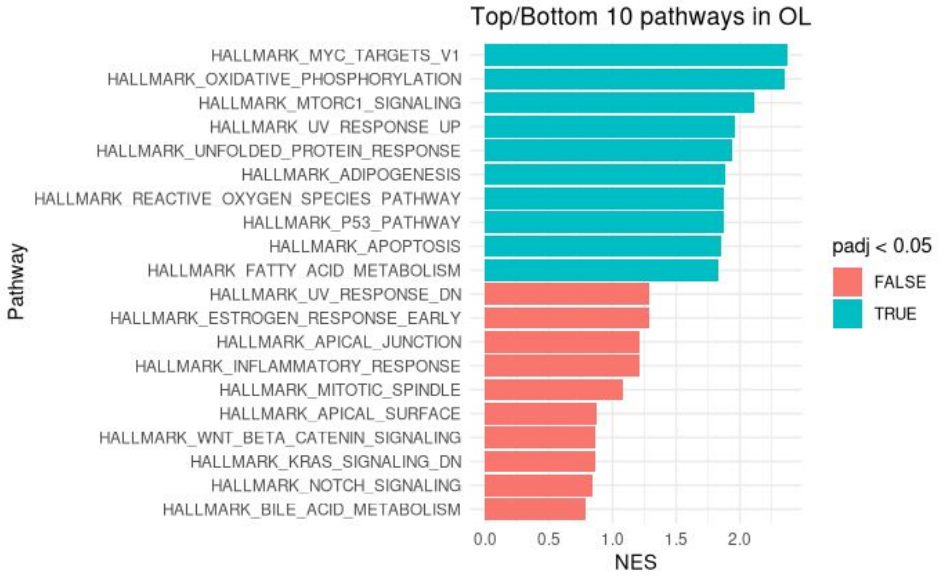


Gene-set enrichment analysis

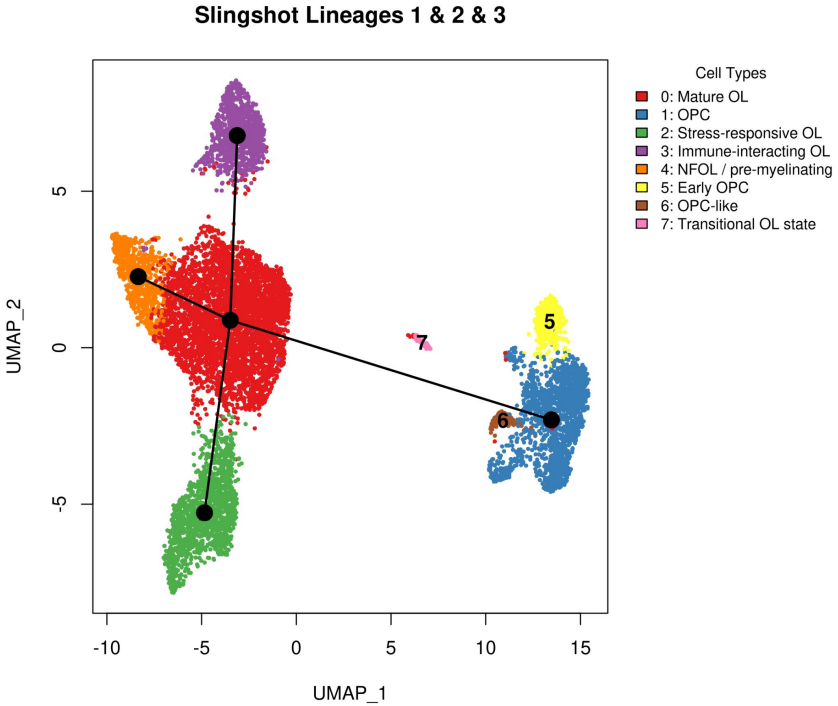
Oligodendrocytes (OL)

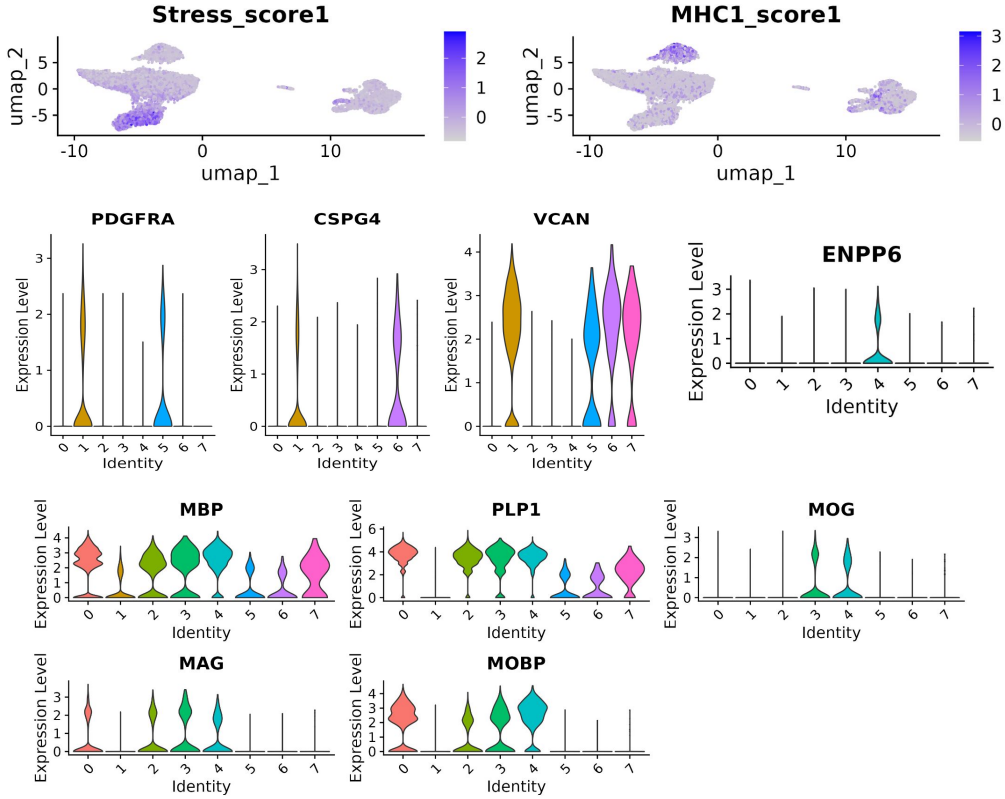
Enriched pathways in MS:

- Oxidative phosphorylation
- MYC targets
- mTORC1 signaling
- Unfolded protein response
- Apoptosis
- Fatty acid metabolism



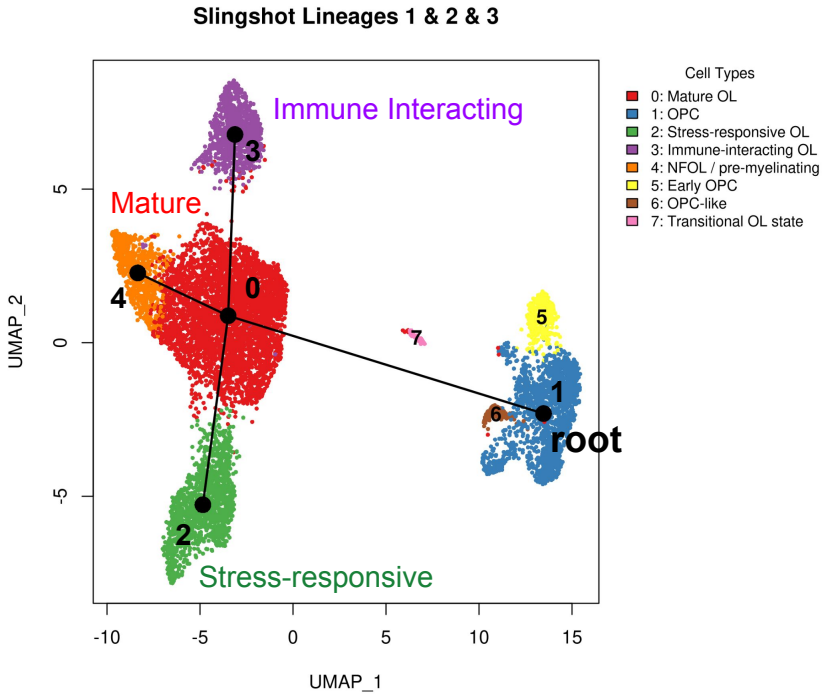
Trajectory Analysis: OL Clustering & Annotation

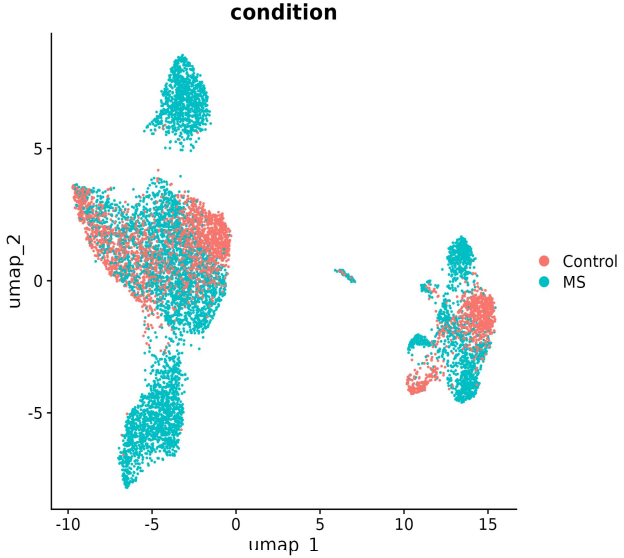


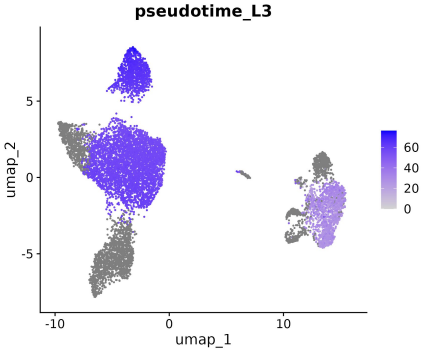


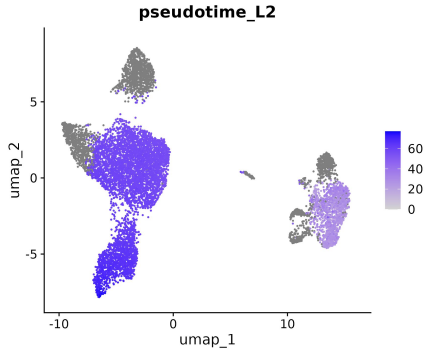
- **Focus on OLs:** Key myelinating cells strongly affected in MS
- **Key result:** OLs span OPC → mature states, with additional stress- and immune-associated trajectories

Slingshot-Based Trajectory Analysis of Oligodendrocytes









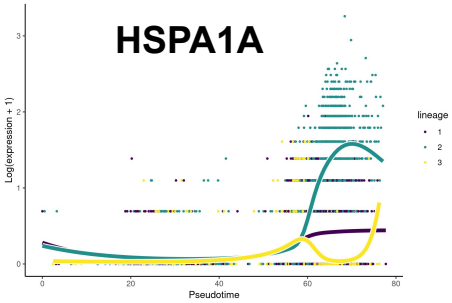
- Lineage 1(1→0→4): OPC → Mature OLs
- Lineage2(1→0→2): OPC → Stress-associated OLs
- Lineage 3(1→0→3): OPC → Immune-interacting OLs

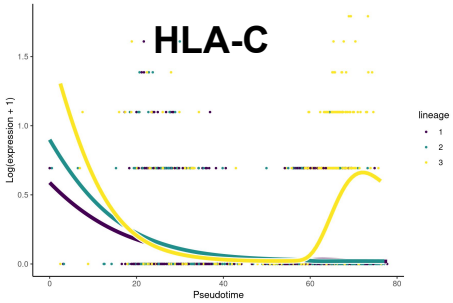
Gene Dynamics Along Disease-Associated Trajectories

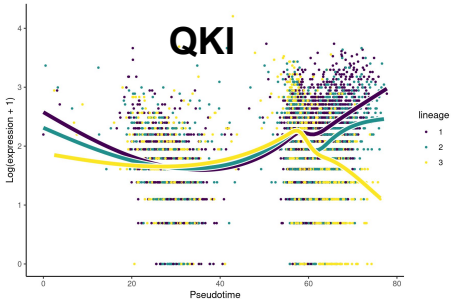
Used **`tradeSeq(fitGAM)`** to model expression changes along pseudotime. Identified:

- **Genes diverging at lineage ends** (`diffEndTest`)
- **Early-changing genes** (`earlyDETest`)

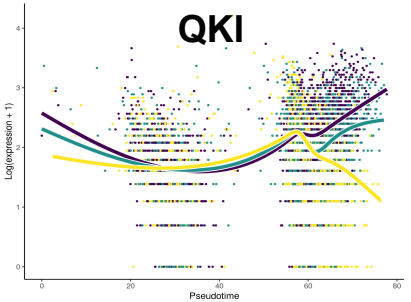
Early pseudotime differences (`earlyDETest`)

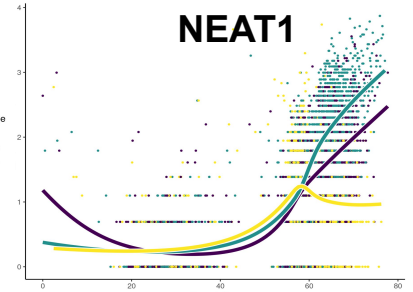






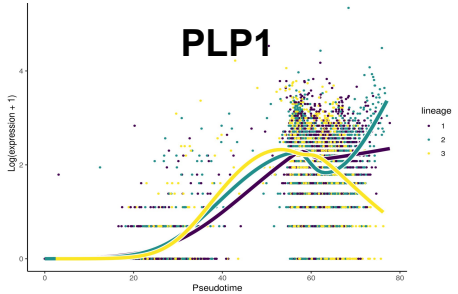
Terminal states (`diffEndTest`)





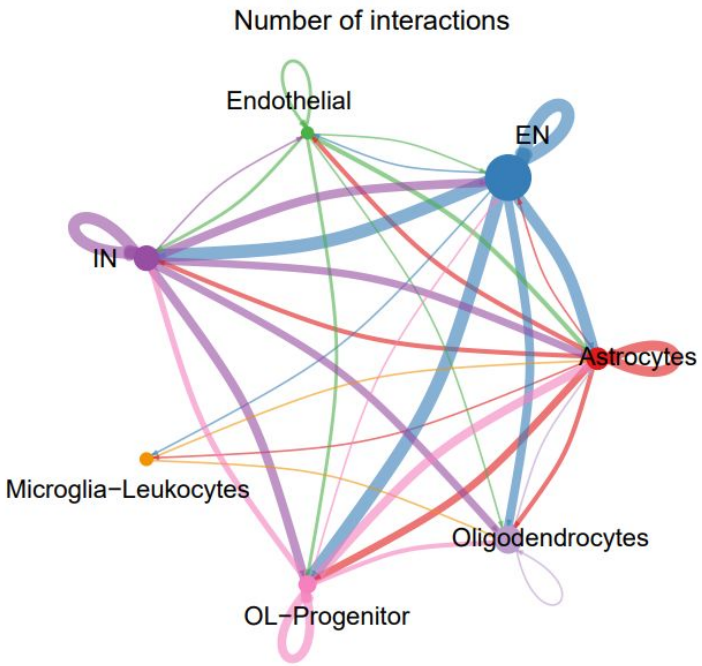
Lineage

- Mature OLs
- Stress-associated OLs
- Immune-interacting OLs

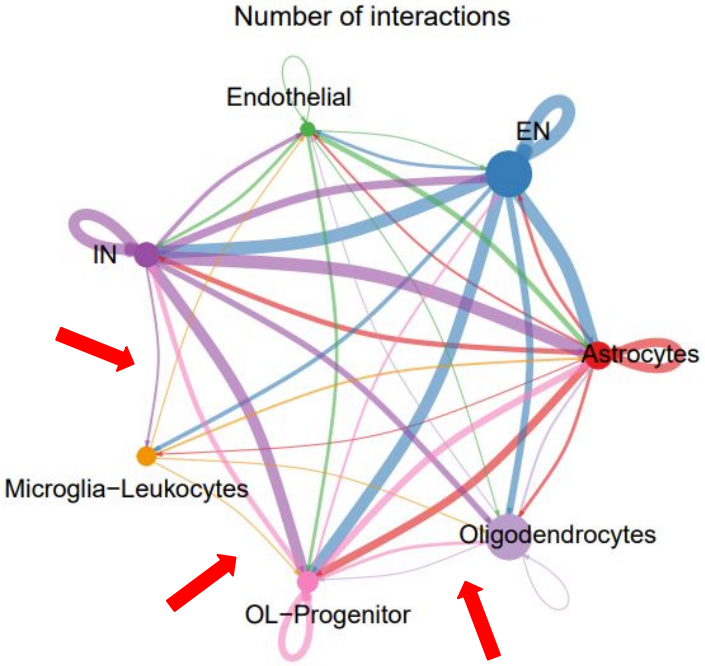


Cell-Cell Communication

Control:

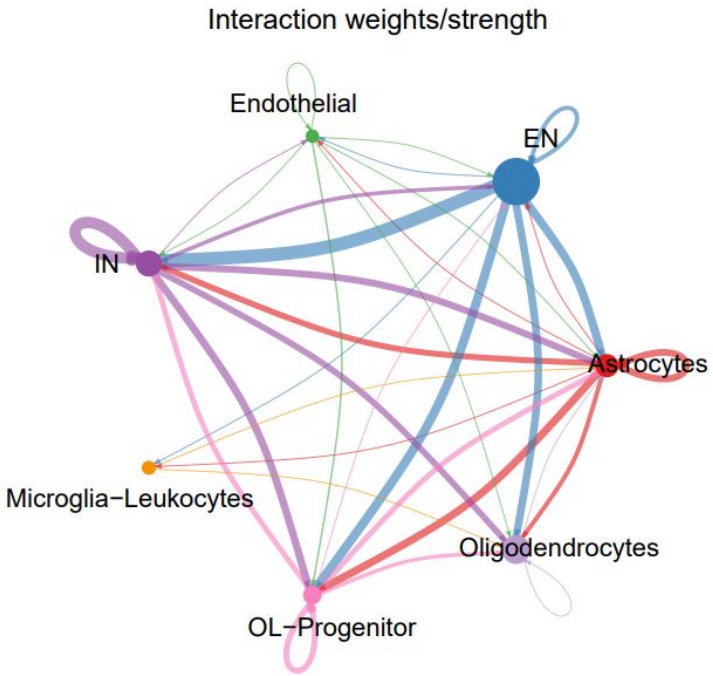


Multiple sclerosis (MS):

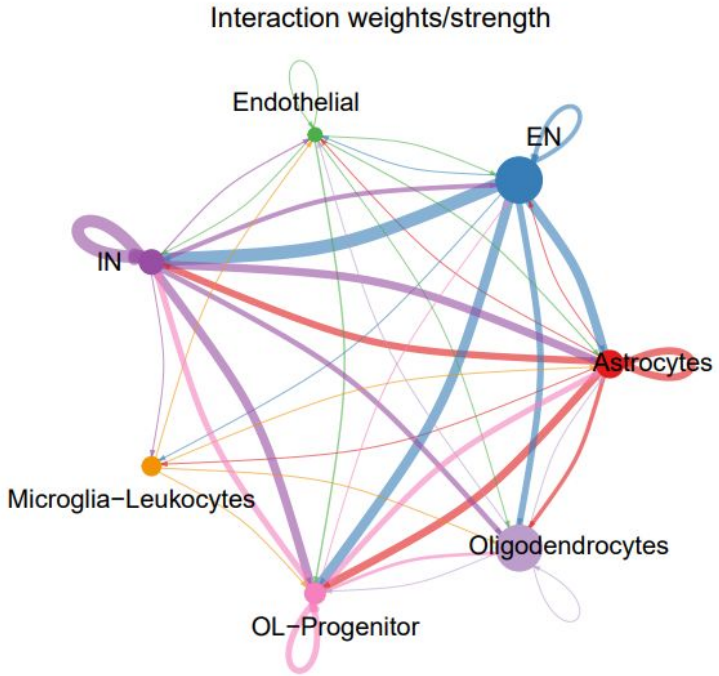


Cell-Cell Communication

Control:



Multiple sclerosis (MS):



Discussion

- Our results align with the neuroinflammatory and neurodegenerative nature of multiple sclerosis, and support previous findings that indicated cell type and state specific disease effects. [1]

- The reference paper reports special susceptibility of a subtype of excitatory neurons in MS, supporting our results in differential composition analysis. Additionally, overrepresentation of microglia was observed, which is a hallmark of MS pathology [1, 2].

Discussion

- Metabolic stress and cell death programs are enriched in OD and ENs, consistent with chronic cellular stress in MS [3].

- Confirmed disease-associated OD trajectories that redirects into two main lineages (stress/immune-related), matching already described OD states in MS and neurodegeneration [4, 5].

- If more time: complete analysis of the dataset, deepen on cell subtypes, spatial transcriptomics…

REFERENCES

[1] Schirmer L et al. Neuronal vulnerability and multilineage diversity in multiple sclerosis. Nature. 2019. PMCID: PMC6731122.

[2] Henstridge, C. M., Tzioras, M., & Paolicelli, R. C. (2019). Glial contribution to excitatory and inhibitory synapse loss in neurodegeneration. Frontiers in Cellular Neuroscience, 13, 63.

[3] Chiricosta, L., Blando, S., D'Angiolini, S., Gugliandolo, A., & Mazzon, E. (2023). A comprehensive exploration of the transcriptomic landscape in multiple sclerosis: A systematic review. International Journal of Molecular Sciences, 24(2), 1448. https://doi.org/10.3390/ijms24021448

[4] Pandey, S., Shen, K., Lee, S.-H., Bohlen, C. J., Yuen, T. J., & Friedman, B. A. (2022). Disease-associated oligodendrocyte responses across neurodegenerative diseases. Cell Reports, 40(5), 111189.

[5] Falcão, A. M., van Bruggen, D., Marques, S., Meijer, M., Jäkel, S., Agirre, E., Floriddia, E. M., Vanichkina, D. P., ffrench-Constant, C., Williams, A., Ortlieb Guerreiro-Cacais, A., & Castelo-Branco, G. (2018). Disease-specific oligodendrocyte lineage cells arise in multiple sclerosis. Nature Medicine, 24, 1837–1844.